100 DAYS OF

DATA ENGINEERING COMPLETE CHALLENGE PLAN

Basic → Intermediate → Job-Ready

Duration	Level	Focus
100 Days	Beginner to Intermediate	Theory + Hands-on Code

Tech Stack

Category	Technology
Language	Python
Database & SQL	PostgreSQL + Advanced SQL
Data Processing	Pandas
Orchestration	Apache Airflow
Transformation	dbt (data build tool)
Big Data	PySpark
Containers	Docker
Cloud	AWS (S3, Glue, Redshift, IAM, Lambda)
Streaming	Apache Kafka (Basics)
Version Control	Git + GitHub

How to Use This Plan

- Har din ka clear objective aur tasks diye gaye hain.
- Theory pehle padho, phir code likho. Notes apni language mein banao.
- GitHub pe daily folder banao (Day-01, Day-02...). README mein progress tracker rakho.
- LinkedIn pe har 3-5 din baad progress ya learning post karo.

- Har phase ke end pe mini project complete karo — yeh portfolio ka hissa banega.
- Agar koi din miss ho jaye to skip mat karo, catch-up karke aage badho.

Challenge Phases Overview

Phase	Days	Focus Area	Outcome
1. Foundations	1–20	Python, SQL, Git, Linux, Pandas	Strong programming + SQL base
2. ETL & Processing	21–40	ETL pipelines, Docker, PostgreSQL	Working data pipelines + Mini Project 1
3. Orchestration	41–60	Airflow, dbt, Data Modeling	Production-style workflows + Mini Project 2
4. Big Data & Cloud	61–80	PySpark + AWS (S3, Glue, Redshift)	Cloud data engineering skills + Mini Project 3
5. Streaming + Capstone	81–100	Kafka basics + End-to-End Project	Complete portfolio project + Job readiness

Daily Time Recommendation

- Theory / Reading: 30–45 minutes
- Hands-on Coding: 60–90 minutes
- Notes + GitHub commit + LinkedIn (optional): 15–20 minutes

Total: ~2 to 2.5 hours per day (adjust according to your schedule)

PHASE 1: Foundations | Days 1–20

Goal: Python, SQL, Git, Linux aur Pandas ki strong foundation banana taake aage ke advanced topics asani se samajh aayein.

Day 01 — Environment Setup & Git Basics

Objective: Development environment ready karo aur Git samjho

Tasks: Python, VS Code/Cursor, Git install karo • GitHub account + pehla repo banao • Basic Git commands (clone, add, commit, push, branch) • Linux terminal basics (cd, ls, mkdir, grep, pipes)

Day 02 — Python Fundamentals Refresh

Objective: Python ki core cheezein revise karo (DE perspective se)

Tasks: Variables, data types, lists, dicts, sets • Functions, *args/**kwargs, lambda • List/dict comprehensions • Error handling (try/except)

Day 03 — Python for Data - File Handling

Objective: Files ke saath kaam karna seekho

Tasks: Read/write text, CSV, JSON • Pathlib module • Working with large files (chunking concept) • Simple log file parser likho

Day 04 — Object Oriented Python (Light)

Objective: Classes aur objects ko practical tarike se samjho

Tasks: Classes, objects, __init__ • Methods vs functions • Simple data pipeline class design • Inheritance ka basic idea

Day 05 — SQL Foundations - SELECT & Filtering

Objective: SQL ki foundation strong karo

Tasks: SELECT, WHERE, ORDER BY, LIMIT • Comparison & logical operators • NULL handling • Practice: 15-20 queries likho

Day 06 — SQL Joins Deep Dive

Objective: Saari important joins master karo

Tasks: INNER, LEFT, RIGHT, FULL OUTER JOIN • Self joins • Multiple table joins • Common join mistakes samjho

Day 07 — SQL Aggregations & GROUP BY

Objective: Data ko summarize karna seekho

Tasks: COUNT, SUM, AVG, MIN, MAX • GROUP BY + HAVING • Multiple aggregations • Practice problems

Day 08 — SQL Subqueries & CTEs

Objective: Complex queries likhna seekho

Tasks: Subqueries (WHERE, FROM, SELECT) • Common Table Expressions (WITH) • CTE vs Subquery • Readable query writing

Day 09 — Window Functions - Part 1

Objective: Window functions ka introduction

Tasks: ROW_NUMBER, RANK, DENSE_RANK • OVER() clause • PARTITION BY • Practical ranking examples

Day 10 — Window Functions - Part 2

Objective: Advanced window functions

Tasks: LAG, LEAD, FIRST_VALUE, LAST_VALUE • Running totals • Moving averages concept • Business use-cases

Day 11 — SQL Practice Day + Data Modeling Intro

Objective: SQL practice + dimensional modeling basics

Tasks: Mixed SQL problems solve karo • OLTP vs OLAP • Star Schema vs Snowflake Schema • Fact & Dimension tables ka idea

Day 12 — Pandas Fundamentals

Objective: Pandas se data handle karna seekho

Tasks: Series & DataFrame • Reading CSV/JSON/Excel • Basic selection, filtering, sorting • head, info, describe, value_counts

Day 13 — Pandas Data Cleaning

Objective: Messy data clean karna seekho

Tasks: Missing values handling • Duplicates remove karna • Data type conversion • String cleaning methods

Day 14 — Pandas Transformations

Objective: Data transform karna

Tasks: apply, map, replace • GroupBy + aggregations • Pivot tables • Merge / Join in Pandas

Day 15 — Pandas + SQL Practice Combo

Objective: Dono tools mix karke practice

Tasks: Same problem SQL aur Pandas dono se solve karo • When to use which tool • Performance basic idea • Notes banao

Day 16 — Linux for Data Engineers

Objective: Server pe kaam karne ke basics

Tasks: File permissions, process management • Cron jobs introduction • SSH basics • Useful commands for DE (find, awk, sed intro)

Day 17 — Python Virtual Environments & Packaging

Objective: Clean project setup seekho

Tasks: venv / virtualenv • requirements.txt • Basic project structure • Logging module introduction

Day 18 — Working with APIs in Python

Objective: API se data nikalna seekho

Tasks: requests library • REST API basics • JSON parsing • Simple API data pull script

Day 19 — Mini Project 0 - Setup & Data Exploration

Objective: Chhota exploration project

Tasks: Public dataset choose karo (Kaggle/data.gov) • Data download + explore with Pandas • Basic cleaning + summary stats • GitHub pe properly document karo

Day 20 — Phase 1 Review + LinkedIn Post

Objective: Foundation solid karo aur share karo

Tasks: Pichle 19 din revise karo • Weak areas identify karo • GitHub README update • LinkedIn pe Phase 1 completion post

PHASE 2: ETL & Data Processing | Days 21–40

Goal: Complete ETL pipelines banana, Docker use karna, aur PostgreSQL ke saath properly kaam karna. End mein ek mini project.

Day 21 — ETL Concepts Deep Dive

Objective: ETL vs ELT samjho aur design principles

Tasks: Extract, Transform, Load stages • Batch vs Incremental • Idempotency concept • Data pipeline design basics

Day 22 — Building Extract Layer (Python)

Objective: Different sources se data nikalna

Tasks: CSV / JSON / API extract scripts • Database extract (psycopg2 / SQLAlchemy) • Error handling in extract • Logging properly add karo

Day 23 — Transform Layer - Business Logic

Objective: Data ko business rules ke mutabiq transform karo

Tasks: Data validation rules • Derived columns • Data type enforcement • Slowly Changing concepts (intro)

Day 24 — Load Layer - Writing to Database

Objective: Clean data database mein load karo

Tasks: INSERT vs UPSERT (ON CONFLICT) • Batch loading • Transaction basics • Simple load script

Day 25 — Full ETL Script - End to End

Objective: Ek complete small ETL banao

Tasks: Extract → Transform → Load script • Config file use karo • Logging + basic error handling • GitHub pe push

Day 26 — Docker Fundamentals

Objective: Containers samjho

Tasks: Docker concepts (image, container, volume) • Dockerfile likhna • docker build & run • Simple Python app containerize karo

Day 27 — Docker Compose & Multi-container

Objective: Multiple services manage karo

Tasks: docker-compose.yml • PostgreSQL container • App + DB together run karo • Volumes for persistence

Day 28 — PostgreSQL Advanced for DE

Objective: Database side skills improve karo

Tasks: Indexes (B-tree basics) • EXPLAIN ANALYZE intro • Vacuum & basic maintenance • Connection pooling concept

Day 29 — Data Quality Basics

Objective: Data ki quality check karna seekho

Tasks: Null checks, range checks, uniqueness • Simple validation framework • Quarantine concept • Great Expectations ka introduction (optional)

Day 30 — Incremental Loading Strategies

Objective: Har baar full load mat karo

Tasks: Timestamp based incremental • ID based incremental • Watermark concept • Practical implementation

Day 31 — Scheduling with Cron + Python

Objective: Pipelines ko schedule karo

Tasks: Cron syntax • Python script schedule karo • Logging to file • Failure notification idea (email/slack concept)

Day 32 — Working with Multiple Data Sources

Objective: Different formats handle karo

Tasks: CSV + API + Database mix • Schema differences handle karo • Unified schema design • Practical multi-source ETL

Day 33 — Error Handling & Retry Logic

Objective: Production-ready error handling

Tasks: Retry strategies • Dead letter concept • Alerting basics • Idempotent pipeline design

Day 34 — Configuration Management

Objective: Hardcoded values mat rakho

Tasks: Config files (YAML/JSON) • Environment variables • Secrets handling basics • Different env (dev/prod) concept

Day 35 — Mini Project 1 - Start

Objective: End-to-end ETL project shuru karo

Tasks: Project idea finalize (e.g. sales / weather / stocks data) • Architecture design • Source + target decide • Repo structure banao

Day 36 — Mini Project 1 - Extract & Transform

Objective: Project ka core logic

Tasks: Extract scripts complete • Transform business rules implement • Unit level testing (manual) • Progress commit

Day 37 — Mini Project 1 - Load + Dockerize

Objective: Project ko containerize karo

Tasks: Load to PostgreSQL • Dockerfile + docker-compose • End-to-end run in containers • README update

Day 38 — Mini Project 1 - Scheduling & Docs

Objective: Project ko schedule aur document karo

Tasks: Cron / simple scheduler • Proper documentation • Sample data + results • GitHub polish

Day 39 — Code Review & Refactoring

Objective: Apne code ko improve karo

Tasks: Code readability improve • Functions modular banao • Comments + docstrings • Performance basic check

Day 40 — Phase 2 Review + LinkedIn

Objective: Phase complete karo aur share karo

Tasks: Phase 2 revise • Mini Project 1 showcase • LinkedIn post with architecture diagram • GitHub project link share

PHASE 3: Orchestration & Modeling | Days 41–60

Goal: Airflow se pipelines orchestrate karna, dbt se transformations manage karna, aur proper data modeling seekhna.

Day 41 — Data Modeling Deep Dive

Objective: Dimensional modeling properly samjho

Tasks: Fact tables (transaction, periodic, accumulating) • Dimension tables (SCD Type 1, 2, 3) • Star vs Snowflake practical differences • Grain concept

Day 42 — Designing a Data Warehouse Schema

Objective: Khud se schema design karo

Tasks: Business process choose karo • Dimensions identify karo • Facts define karo • ER diagram / schema draw karo

Day 43 — Apache Airflow Introduction

Objective: Airflow ka architecture samjho

Tasks: DAG, Task, Operator concepts • Airflow architecture (Scheduler, Executor, Webserver) • Installation (Local / Docker) • First simple DAG

Day 44 — Airflow Operators & Sensors

Objective: Common operators use karo

Tasks: PythonOperator, BashOperator • PostgresOperator • Sensors (FileSensor, TimeSensor) • Task dependencies (>> and <<)

Day 45 — Airflow Advanced - XComs & Variables

Objective: Tasks ke beech data share karo

Tasks: XComs (push/pull) • Variables & Connections • Jinja templating basics • Practical example

Day 46 — Building ETL DAG in Airflow

Objective: Pehle wale ETL ko Airflow mein lao

Tasks: Extract-Transform-Load as separate tasks • Proper retries & timeouts • Email alerts on failure • DAG documentation

Day 47 — Airflow Best Practices

Objective: Production style DAGs likho

Tasks: Idempotent tasks • Avoid top-level code • Task granularity • Testing DAGs locally

Day 48 — dbt Introduction

Objective: dbt kya hai aur kyun use hota hai

Tasks: dbt concepts (models, sources, tests) • dbt project structure • Installation + first model • ref() and source()

Day 49 — dbt Models & Materializations

Objective: Different materializations samjho

Tasks: Table, View, Incremental, Ephemeral • When to use which • Incremental models practical • Schema.yml

Day 50 — dbt Tests & Documentation

Objective: Data tests aur docs

Tasks: Generic tests (unique, not_null, relationships) • Singular tests • dbt docs generate • Source freshness

Day 51 — dbt + Airflow Integration

Objective: Dono tools ko jodna

Tasks: dbt run from Airflow • Cosmos / simple BashOperator approach • Full transformation pipeline • Practical setup

Day 52 — Slowly Changing Dimensions Implementation

Objective: SCD Type 2 practically implement karo

Tasks: SCD Type 1 vs Type 2 • Python / SQL se SCD2 • dbt snapshot (intro) • Historical tracking

Day 53 — Data Lineage & Catalog Basics

Objective: Data kahan se aaya yeh track karo

Tasks: Lineage concept • Simple documentation approach • Why it matters in teams • Tools overview (optional)

Day 54 — Mini Project 2 - Design

Objective: Orchestrated pipeline project design

Tasks: Project idea (e.g. e-commerce analytics) • Schema design (star schema) • Airflow + dbt architecture • Repo setup

Day 55 — Mini Project 2 - Development

Objective: Core development

Tasks: dbt models likho • Airflow DAG banao • Tests add karo • Local end-to-end run

Day 56 — Mini Project 2 - Polish

Objective: Project ko production-like banao

Tasks: Error handling improve • Documentation complete • Sample dashboard idea (optional) • GitHub ready

Day 57 — Testing Data Pipelines

Objective: Pipelines ko test karna seekho

Tasks: Unit tests for transforms • Data quality tests • DAG testing approaches • CI idea (GitHub Actions intro)

Day 58 — Monitoring Basics for Pipelines

Objective: Pipelines ki health check

Tasks: Logging best practices • Simple metrics (rows processed, duration) • Alerting strategies • What to monitor

Day 59 — Code Organization & Modular Design

Objective: Bade projects ka structure

Tasks: Project folder structure • Reusable components • Config management advanced • Documentation standards

Day 60 — Phase 3 Review + LinkedIn

Objective: Phase complete + share

Tasks: Full phase revise • Mini Project 2 showcase • Architecture diagram post • LinkedIn + GitHub update

PHASE 4: Big Data & AWS Cloud | Days 61–80

Goal: PySpark se large data process karna aur AWS services (S3, Glue, Redshift) ke saath real cloud data engineering skills banana.

Day 61 — Big Data Concepts

Objective: Big Data ecosystem samjho

Tasks: Volume, Velocity, Variety • Hadoop ecosystem overview • When to use Spark • Distributed computing basics

Day 62 — PySpark Setup & RDDs

Objective: Spark environment + RDD basics

Tasks: Local Spark setup / Docker • RDD concepts • Transformations vs Actions • Simple RDD operations

Day 63 — PySpark DataFrames

Objective: DataFrame API master karo

Tasks: Creating DataFrames • Select, filter, groupBy • Joins in Spark • Schema management

Day 64 — PySpark Advanced Transformations

Objective: Complex transformations

Tasks: Window functions in Spark • UDFs (careful use) • Handling nulls & duplicates • Performance tips (intro)

Day 65 — Spark SQL & Catalog

Objective: SQL interface of Spark

Tasks: Spark SQL • Temporary views • Catalog operations • Mixed DataFrame + SQL

Day 66 — AWS Account & IAM Basics

Objective: AWS security foundation

Tasks: AWS Free Tier account • IAM users, roles, policies • Least privilege principle • CLI setup (aws configure)

Day 67 — Amazon S3 Deep Dive

Objective: Object storage master karo

Tasks: Buckets, objects, prefixes • Storage classes • Versioning, lifecycle • Python (boto3) se S3 operations

Day 68 — AWS Glue Introduction

Objective: Serverless ETL service

Tasks: Glue Data Catalog • Crawlers • Glue Jobs (Spark) • Simple Glue job create karo

Day 69 — AWS Glue Practical

Objective: Glue se practical ETL

Tasks: S3 se data read/write • Transformations in Glue • Job bookmarks (incremental) • Monitoring Glue jobs

Day 70 — Amazon Redshift Basics

Objective: Cloud data warehouse

Tasks: Redshift architecture (leader + compute) • Distribution styles & sort keys • CREATE TABLE best practices • Loading data (COPY command)

Day 71 — Redshift + S3 Integration

Objective: S3 se Redshift load karo

Tasks: COPY from S3 • Manifest files • Compression & performance • Basic query optimization

Day 72 — AWS Lambda for Data

Objective: Serverless functions for light tasks

Tasks: Lambda basics • S3 event triggers • Simple data validation Lambda • Limitations samjho

Day 73 — Building Cloud ETL Pipeline

Objective: S3 → Glue → Redshift flow

Tasks: Architecture design • End-to-end pipeline • Error handling • Cost awareness

Day 74 — PySpark on AWS (EMR / Glue)

Objective: Spark cloud pe chalana

Tasks: EMR vs Glue comparison • Submitting Spark jobs • Reading from S3 in Spark • Writing results back

Day 75 — Data Lake Concepts on AWS

Objective: Data Lake architecture

Tasks: Data Lake vs Warehouse • Bronze / Silver / Gold (Medallion) • S3 as Data Lake • Catalog importance

Day 76 — Mini Project 3 - Design

Objective: Cloud data pipeline project

Tasks: Project idea finalize • Architecture (S3 + Glue + Redshift) • Data sources decide • Cost estimation (Free Tier)

Day 77 — Mini Project 3 - Implementation

Objective: Core cloud pipeline

Tasks: S3 setup + data landing • Glue jobs / Spark transforms • Redshift tables + load • Basic validation

Day 78 — Mini Project 3 - Orchestration

Objective: Airflow / Step Functions se control

Tasks: Pipeline orchestration • Failure handling • Documentation • End-to-end test

Day 79 — Cost Optimization & Best Practices

Objective: Cloud pe paise bachana

Tasks: S3 storage optimization • Glue DPU understanding • Redshift pause/resume • Monitoring costs

Day 80 — Phase 4 Review + LinkedIn

Objective: Phase complete + strong showcase

Tasks: Full revise • Architecture diagram • LinkedIn detailed post • GitHub project polish

PHASE 5: Streaming + Capstone Project | Days 81–100

Goal: Kafka basics seekhna aur ek complete end-to-end Data Engineering project banana jo portfolio mein dikhaya ja sake.

Day 81 — Streaming Concepts

Objective: Batch vs Streaming samjho

Tasks: Real-time vs near real-time • Event driven architecture • Use cases (fraud, monitoring, recommendations) • Challenges in streaming

Day 82 — Apache Kafka Fundamentals

Objective: Kafka core concepts

Tasks: Topics, Partitions, Offsets • Producers & Consumers • Consumer Groups • Brokers & Zookeeper/KRaft intro

Day 83 — Kafka Hands-on (Local)

Objective: Practically Kafka use karo

Tasks: Local Kafka setup (Docker) • Create topic • Python producer (kafka-python) • Python consumer

Day 84 — Kafka + Python Integration

Objective: Real example banao

Tasks: Produce JSON events • Consume & process • Simple transformation on stream • Error handling

Day 85 — Kafka Design Best Practices

Objective: Production considerations

Tasks: Partitioning strategy • Replication factor • Message keys • Exactly-once vs at-least-once

Day 86 — Capstone Project - Ideation

Objective: Final project decide karo

Tasks: Project idea brainstorm (e.g. real-time analytics, end-to-end batch+stream) • Requirements define • Architecture high-level • Tech stack finalize

Day 87 — Capstone - Architecture Design

Objective: Detailed design

Tasks: Data sources • Ingestion layer • Processing (batch + stream) • Storage + Serving layer • Draw architecture diagram

Day 88 — Capstone - Data Ingestion

Objective: Ingestion layer build karo

Tasks: Batch sources setup • Streaming source (Kafka) • Landing zone (S3) • Initial validation

Day 89 — Capstone - Processing Layer

Objective: Transforms implement karo

Tasks: Spark / Glue jobs • dbt models (if applicable) • Business logic • Data quality checks

Day 90 — Capstone - Storage & Modeling

Objective: Warehouse / Lake modeling

Tasks: Redshift / curated zone tables • Dimensional model • SCD handling • Optimization

Day 91 — Capstone - Orchestration

Objective: Poori pipeline control karo

Tasks: Airflow DAGs • Dependencies • Retries & alerts • End-to-end scheduling

Day 92 — Capstone - Streaming Part

Objective: Real-time component

Tasks: Kafka integration • Stream processing • Near real-time updates • Testing stream path

Day 93 — Capstone - Testing & Data Quality

Objective: Quality ensure karo

Tasks: End-to-end tests • Data quality suite • Edge cases • Performance basic check

Day 94 — Capstone - Documentation

Objective: Professional documentation

Tasks: Architecture decision records • How to run guide • Data dictionary • Assumptions & limitations

Day 95 — Capstone - Polish & Demo Prep

Objective: Presentation ready banao

Tasks: README professional • Demo script • Screenshots / GIFs • Known issues list

Day 96 — System Design for Data Engineers

Objective: Interview style thinking

Tasks: How to design a data platform • Trade-offs (cost, latency, consistency) • Common patterns • Practice explaining architecture

Day 97 — Optimization Techniques

Objective: Pipelines ko better banao

Tasks: Spark optimization tips • SQL optimization • File formats (Parquet, ORC) • Partitioning strategies

Day 98 — CI/CD Basics for Data

Objective: Pipelines ko automate deploy karo

Tasks: GitHub Actions intro • Simple CI for dbt / Python • Environment promotion concept • Infrastructure as code idea

Day 99 — Portfolio + Resume Prep

Objective: Job ready materials

Tasks: GitHub profile polish • Project READMEs improve • LinkedIn featured section • Resume bullet points for DE

Day 100 — Final Review + Celebration Post

Objective: Challenge complete!

Tasks: 100 days revise highlights • Biggest learnings • Final LinkedIn post with full journey • Next goals set karo (job applications / advanced topics)

GitHub Repository Structure (Recommended)

```
100-Days-of-Data-Engineering/
■■■ README.md                ← Challenge overview + progress tracker
■■■ Day-01/
■   ■■■ README.md            ← Objective + tasks + notes
■   ■■■ theory.md
■   ■■■ code/
■■■ Day-02/
■   ■■■ ...
■■■ projects/
■   ■■■ mini-project-1/
■   ■■■ mini-project-2/
■   ■■■ mini-project-3/
■   ■■■ capstone-project/
■■■ resources/
■   ■■■ useful-links.md
■■■ templates/
    ■■■ daily-readme-template.md
```

LinkedIn Strategy Tips

- Har din post mat karo. Har 3–5 din baad value-added post best hai.
- Post types: Progress update, Concept explanation, Mistake/learning, Mini-project showcase.
- Code screenshot ya architecture diagram zaroor dalo — engagement badhta hai.
- Hashtags: #100DaysOfDataEngineering #DataEngineering #LearnInPublic #Python #AWS #Airflow
- Comments ka reply do — network build hota hai.

Success Tips

- Consistency > Intensity. Roz thoda behtar hai week mein ek din zyada karne se.
- Notes apni language mein likho. Baad mein revise karte waqt kaam aayenge.
- Jab stuck ho to 30-40 minute try karo, phir help lo (docs, StackOverflow, ya mujhse).
- Har mini project ko GitHub pe achhe se document karo — yeh interviews mein kaam aayega.
- Phase ke end pe LinkedIn pe zaroor share karo. Public accountability help karti hai.

Yeh plan tumhari consistency aur direction ke liye hai.
Rozana progress ke saath aage badhte raho. Best of luck!

— 100 Days of Data Engineering Challenge —